

QUESTIONS 31 / 31 completed

Shpere ▾

SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.*;
2 class sphere{
3     public static void main(String[] args){
4         Scanner sc= new Scanner(System.in);
5         int r;
6         r=sc.nextInt();
7         double sphere;
8         sphere=(4/3)*Math.PI*r*r*r;
9         System.out.print("Volume of the Sphere: "+sphere);
10    }
11 }
```

OUTPUT

```
Volume of the Sphere: 1077.566280181299
```

INPUT

7

QUESTIONS 31 / 31 completed

Palindrome ▾

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String s = sc.nextLine().toLowerCase().replaceAll("\\s+", "");
6         boolean isPalindrome = true;
7         for (int i = 0; i < s.length() / 2; i++) {
8             if (s.charAt(i) != s.charAt(s.length() - 1 - i)) {
9                 isPalindrome = false;
10                break;
11            }
12        }
13        System.out.println("OUTPUT: " + (isPalindrome ? "Palindrome" : "Not Palindrome"));
14        sc.close();
15    }
16 }
```

OUTPUT

OUTPUT: Palindrome

INPUT

elle

QUESTIONS 31 / 31 completed

Armstrong ▾

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.Scanner;
2 class armstrong{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int num=sc.nextInt();
6         int result=0,rem;
7         int n=num;
8         while(n!=0){
9             rem=n%10;
10            result+=rem*rem*rem;
11            n/=10;
12        }
13        if(num==result){
14            System.out.print(num+" is an Armstrong number");
15        }
16        else{
17            System.out.print(num+" is not an Armstrong number");
18        }
19    }
20 }
```

OUTPUT

```
153 is an Armstrong number
```

INPUT

153

QUESTIONS 31 / 31 completed

RT-num.Pattern ▾

**RT-
NUM.PATTERN**

Write a Java program to
print the following
pattern using loops:1

22
333
4444
55555

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.Scanner;
2 class numpattern{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n=sc.nextInt();
6         for(int i=1;i<=n;i++){
7             for(int j=1;j<=i;j++){
8                 System.out.print(i);
9             }
10            System.out.println();
11        }
12    }
13 }
```

OUTPUT

```
1
22
333
4444
55555
```

INPUT

5

QUESTIONS 31 / 31 completed

Perfect

PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class PerfectNumber {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int sum = 0;
7         for (int i = 1; i <= n / 2; i++) {
8             if (n % i == 0) {
9                 sum += i;
10            }
11        }
12        if (sum == n) {
13            System.out.println("Perfect");
14        } else {
15            System.out.println("Not Perfect");
16        }
17        sc.close();
18    }
19 }
```

OUTPUT

Perfect

INPUT

6

QUESTIONS 31 / 31 completed

Max Array ▾

MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.Scanner;
2 public class MaxArray {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] arr = new int[n];
7         for (int i = 0; i < n; i++) {
8             arr[i] = sc.nextInt();
9         }
10        int max = arr[0];
11        for (int i = 1; i < n; i++) {
12            if (arr[i] > max) {
13                max = arr[i];
14            }
15        }
16        System.out.println("Largest element = " + max);
17    }
18 }
```

OUTPUT

```
Largest element = 9
```

INPUT

```
5
9
3
8
7
6
```

QUESTIONS 31 / 31 completed

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class SecondMax {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         if (!sc.hasNextInt()) return;
6         int n = sc.nextInt();
7         if (n < 2) {
8             System.out.println("Array must have at least two elements");
9             return;
10        }
11        int max = Integer.MIN_VALUE;
12        int secondMax = Integer.MIN_VALUE;
13        for (int i = 0; i < n; i++) {
14            int current = sc.nextInt();
15            if (current > max) {
16                secondMax = max;
17                max = current;
18            } else if (current > secondMax && current < max) {
19                secondMax = current;
20            }
21        }
```

OUTPUT

```
Second Largest : 6
```

INPUT

```
5
3
4
6
7
2
```

QUESTIONS 31 / 31 completed

Min

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class SmallestElement {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] arr = new int[n];
7         for (int i = 0; i < n; i++) {
8             arr[i] = sc.nextInt();
9         }
10        int smallest = arr[0];
11        for (int i = 1; i < n; i++) {
12            if (arr[i] < smallest) {
13                smallest = arr[i];
14            }
15        }
16        System.out.println("Smallest element is: " + smallest);
17        sc.close();
18    }
19 }
```

OUTPUT

```
Smallest element is: 1
```

INPUT

```
5
3
7
6
2
1
```


QUESTIONS 31 / 31 completed

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class SecondSmallest {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] arr = new int[n];
7         for (int i = 0; i < n; i++) {
8             arr[i] = sc.nextInt();
9         }
10        int firstSmallest = Integer.MAX_VALUE;
11        int secondSmallest = Integer.MAX_VALUE;
12        for (int i = 0; i < n; i++) {
13            int val = arr[i];
14            if (val < firstSmallest) {
15                secondSmallest = firstSmallest;
16                firstSmallest = val;
17            } else if (val > firstSmallest && val < secondSmallest) {
18                secondSmallest = val;
19            }
20        }
21        if (secondSmallest == Integer.MAX_VALUE) {
```

OUTPUT

```
Second smallest element: 2
```

INPUT

```
5
1
2
3
4
5
```

QUESTIONS 31 / 31 completed

Sort-B ▾

SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] arr = new int[n];
7
8         for (int i = 0; i < n; i++) {
9             arr[i] = sc.nextInt();
10        }
11        for (int i = 0; i < n - 1; i++) {
12            for (int j = 0; j < n - 1 - i; j++) {
13                if (arr[j] > arr[j + 1]) {
14                    int temp = arr[j];
15                    arr[j] = arr[j + 1];
16                    arr[j + 1] = temp;
17                }
18            }
19        }
20        for (int i = 0; i < n; i++) {
21            System.out.print(arr[i]);
```

OUTPUT

```
1 2 3 5 6
```

INPUT

```
5
2
6
1
3
5
```

QUESTIONS 31 / 31 completed

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class InsertionSort {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] arr = new int[n];
7         for (int i = 0; i < n; i++) {
8             arr[i] = sc.nextInt();
9         }
10        for (int i = 1; i < n; i++) {
11            int key = arr[i];
12            int j = i - 1;
13            while (j >= 0 && arr[j] > key) {
14                arr[j + 1] = arr[j];
15                j = j - 1;
16            }
17            arr[j + 1] = key;
18        }
19        System.out.println("Sorted array:");
20        for (int i = 0; i < n; i++) {
21            System.out.print(arr[i] + " ");
```

OUTPUT

```
Sorted array:
2 3 5 6 8
```

INPUT

```
5
6
8
3
2
5
```